

FloodGuard AI — Technical Report

ML Opsidian: Genesis — Final Round Submission

June 2026

Executive Summary

FloodGuard AI is an end-to-end MLOps system that turns the Initial Round flood-risk model for Sri Lanka into a deployed product. It packages a 223-feature, 3-model ensemble (the production evolution of the Initial Round's winning ideas) behind a FastAPI service that serves a prediction form, a REST API, an AI-generated risk report (Claude, with a deterministic fallback), and a live monitoring dashboard — all from a single Docker image deployed on Render.

This report covers Problem Understanding, System Architecture, the Machine Learning Approach (Initial Round summary and Final Round improvements), and MLOps Practices (deployment, monitoring, version control), per the Final Round deliverable requirements.

1. Problem Understanding

1.1 Understanding of the Challenge

The underlying task, inherited from the Initial Round, is to predict a continuous `flood_risk_score` in `[0, 1]` for locations across Sri Lanka's 25 districts, given a synthetic dataset of 20,886 training records and 5,300 test records with 47 columns covering:

- **Geography:** latitude, longitude, district, elevation, distance to river
- **Climate:** 7-day and monthly rainfall, drainage index, seasonal index
- **Land:** land cover, soil type, NDVI/NDWI vegetation/water indices, built-up %
- **Infrastructure & socioeconomics:** population density, infrastructure score, road/water/electricity quality, distance to hospital/evacuation point
- **Event context:** current flood occurrence, inundated area, historical flood count, "is this place good to live" flag and reason

Models are scored with a **custom metric** that combines absolute and squared error with an explained-variance penalty:

$$\text{score} = (0.5 * \text{MAE} + 0.5 * \text{RMSE}) * (1 + \max(0, 1 - \text{explained_variance}))$$

Lower is better. The $(1 + \max(0, 1 - \text{EV}))$ term means a model that fits the *absolute level* of the target well but fails to explain its *variance* (e.g. a model that always predicts the mean) is penalized — both calibration and discrimination matter.

The Final Round reframes this from "get the best leaderboard score" to: **take that model and turn it into a usable, deployed, monitored system** while keeping the Initial Round model as "a significant component" of the solution (Rule 1).

1.2 Key Observations

A few properties of the dataset shaped both the Initial Round model and how it had to be re-engineered for production:

1. **The data is synthetic and noisy.** The target correlates only moderately with any single feature; no individual column dominates. This is why the Initial Round's strongest models were **ensembles of gradient-boosted trees with district-level aggregate features**, not simple linear models.
2. **district is the single most informative feature.** Across every model version, **district target-encoding (`district_te`)** is the top feature by gain — by a wide margin (gain $\approx 5,345$ vs. $\approx 2,340$ for the next-ranked feature in the Final Round model). Sri Lanka's flood risk is strongly regional, so district-level statistics (mean/std/quantiles/ranks of rainfall, elevation, drainage, etc.) are highly predictive.
3. **Several numeric columns contain out-of-range/negative synthetic values** (e.g. ~ 100 - 125 rows each with negative `rainfall_7d_mm`, `monthly_rainfall_mm`, `distance_to_river_m` out of 20,886). `log1p()` of a negative value is `NaN` by definition — this is **inherited, expected behavior**, not a bug, and

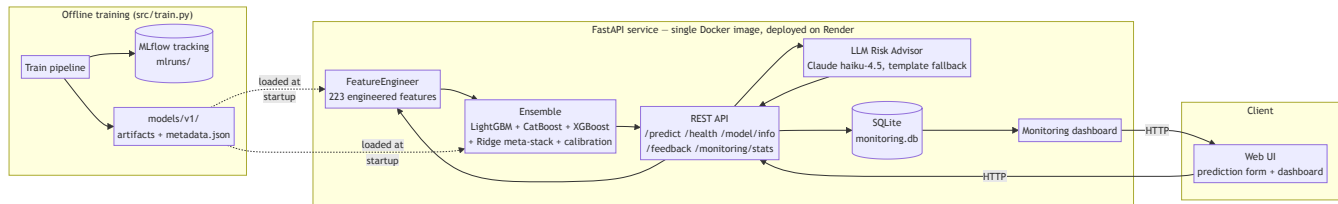
LightGBM/CatBoost/XGBoost natively handle NaN as a "missing value" split direction.

src/data_validation.py flags (but does not reject) such values, both at training time and per inference request (flag_out_of_distribution).

4. **Calibration matters as much as discrimination.** The Initial Round's best submission had a predicted-score mean (0.4780) that matched the **training target mean (0.4780) almost exactly**, while an earlier version (v9) was ~0.006 lower and scored worse on the custom metric — directly because of the $(1 + \max(0, 1 - EV))$ term's sensitivity to systematic bias. The Final Round model **explicitly computes and applies a mean-calibration shift** as a final post-processing step (§3.2).
 5. **A model trained for batch leaderboard scoring is not a model that can score one new record on demand.** The Initial Round pipeline computes district statistics, KMeans clusters, KNN target-statistics, and target encodings *over the whole training set at once* — fine for a CSV-in/CSV-out script, but not directly usable for "a user fills out a form and gets one prediction back in milliseconds." Re-architecting this into a **fit/transform FeatureEngineer** that can be fit once and applied to a single new row was the central engineering problem of the Final Round (§3.2).
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2. System Architecture

2.1 Overall Architecture Diagram



2.2 Workflow Explanation

Offline (training) path:

1. `src/train.py` loads `data/train.csv`, runs `validate_training_data()` (`src/data_validation.py`) to sanity-check schema, nulls, and value ranges.
2. `FeatureEngineer.fit_transform()` (`src/features.py`) builds the 223-column feature matrix and **fits and stores** every stateful component needed to transform a *single future row* the same way (district lookup tables, KMeans models, a `BallTree` for KNN target statistics, target-encoding maps, and empirical quantile mappings).
3. A 5-fold CV loop trains **LightGBM, CatBoost, and XGBoost** on each fold, producing out-of-fold (OOF) predictions for all three.
4. A **Ridge meta-learner** (non-negative weights) is fit on the OOF predictions to produce a stacked ensemble prediction (the Initial Round's winning idea).
5. A **calibration shift** (`train_target_mean - stacked_oof_mean`) is computed and will be added to every future prediction, then clipped to `[0, 1]`.
6. All of this — hyperparameters, per-fold metrics, OOF metrics, the calibration shift, and the `metadata.json` artifact — is **logged to MLflow** (`mlruns/`) for experiment tracking and run comparison.
7. Artifacts (`feature_engineer.joblib`, the three model lists, the Ridge model, and `metadata.json`) are saved to `models/v1/`.

Online (serving) path:

1. On startup, `app/main.py`'s lifespan loads a `FloodRiskPredictor` (`src/inference.py`), which deserializes every artifact in `models/v1/`.
2. A user submits the prediction form (or calls `POST /predict` directly). The request is validated against a pydantic schema (`app/schemas.py`).
3. `FeatureEngineer.transform()` turns the single record into the same 223-column feature vector used in training — filling any optional fields (composite indices, "current event" fields, `*_qmap` columns) using the fitted defaults/mappings from training.
4. The three base models predict; the Ridge meta-model stacks them; the stored calibration shift is applied and the result clipped to `[0, 1]`.
5. `src/data_validation.flag_out_of_distribution()` checks the raw inputs against expected ranges and attaches any warnings.

6. The top contributing features (precomputed LightGBM gain importances) are attached, and `src/llm_advisor.generate_risk_report()` produces a plain-language summary + recommendations (Claude, or a template fallback).
 7. The full prediction (inputs, score, category, latency, OOD flags) is logged to **SQLite** (`app/monitoring.py`) and the response returned to the client.
 8. `/monitoring/stats` aggregates the SQLite log for the `/dashboard` page (score distribution, category/district breakdowns, latency, user feedback).
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3. Machine Learning Approach

3.1 Initial Round Model Summary

The Initial Round solution (`solution_v10.py`) was iterated through ten versions (`solution_v2.py` ... `solution_v10.py`), converging on:

- **5-model ensemble**: LightGBM, LightGBM-DART, CatBoost, XGBoost, and `ExtraTreesRegressor` (Optuna-tuned).
- **~150-column feature matrix** built from: date/seasonal features, geo features, **district-level mean/std/quantile/rank statistics** for ~14 numeric columns, **KMeans clusters** (k = 5, 10, 20), **haversine-KNN target statistics** (k = 5...100) via a `BallTree` , **target encoding** for categorical columns, label encoding, and interaction features (e.g. extreme-weather × terrain-roughness).
- **10-fold × 3-seed cross-validation**, with a **Ridge meta-learner** stacking the five models' OOF predictions (replacing an earlier scipy-optimize blend).
- **10 rounds of pseudo-labeling** on the test set to squeeze out additional signal.
- **Post-processing**: shifting the final prediction mean to exactly match the training target mean — the calibration insight described in §1.2.

This pipeline produced the team's best Initial Round submission (custom metric ≈ **0.383**). It is powerful, but every one of its strengths is also why it **cannot be served as-is**: pseudo-labeling requires the entire test set up front; 10-fold × 3-seed training takes tens of minutes; and the district/KMeans/ KNN/target-encoding statistics are computed in one shot over the whole dataset, with no mechanism to apply them to a brand-new single row.

3.2 Improvements Made

The Final Round model (`src/train.py` , `src/features.py` , `src/inference.py` , version **v1**) is a **restructuring, not a replacement**, of the Initial Round approach — Rule 1 ("the Initial Round work must remain a significant component") is satisfied by carrying forward its exact feature-engineering philosophy and its two highest-leverage ideas (ensemble + Ridge stacking, and mean calibration), while making three changes that make it servable:

1. **FeatureEngineer as a fit/transform/save/load object**. Every "compute over the whole dataset" step in v10 was rewritten as *fit on training data → store a small lookup/model artifact → apply to any new row*:

v10 (batch)	FloodGuard AI (FeatureEngineer)
District stats computed via <code>groupby</code> over all rows	Precomputed per-district lookup table, joined onto new rows
KMeans fit and <code>.predict()</code> on the same data	KMeans models fit once, saved, <code>.predict()</code> on the new row
KNN target stats via <code>BallTree</code> over all rows	Same <code>BallTree</code> , built once at fit time, queried per new row
Target encoding via K-fold OOF means	OOF means for training; full-train means for new rows
<code>*_qmap</code> columns (no raw counterpart)	Empirical monotonic map learned at fit time, <code>np.interp</code> at transform time

v10 (batch)	FloodGuard AI (FeatureEngineer)
Composite indices (<code>seasonal_index</code> , etc.) assumed present	Default to the district mean (or global mean) if the user omits them

The result is a 223-feature matrix that is **column-for-column identical** whether produced for 20,886 training rows or for 1 new row submitted through the API — verified by `tests/test_features.py` .

- 2. **Reduced, fixed-hyperparameter ensemble.** LightGBM-DART and ExtraTrees were dropped (in v10 they contributed marginal gain over LightGBM/CatBoost/XGBoost while roughly doubling artifact count and serialization complexity), 10-fold × 3-seed CV was reduced to **5-fold**, and pseudo-labeling was dropped entirely (it is fundamentally a batch/test-set technique with no analogue for "score one new record"). Hyperparameters were fixed to sensible, previously-tuned values rather than re-running Optuna, trading a small amount of leaderboard score for a **~45-second, fully reproducible training run**.
- 3. **The same calibration step, computed and stored explicitly.** `src/train.py` computes `calibration_shift = train_target_mean - stacked_oof_mean` and stores it in `metadata.json` ; `FloodRiskPredictor` adds it to every prediction and clips to `[0, 1]` .

Resulting model (v1) performance

Metric	LightGBM	CatBoost	XGBoost	Ridge-stacked	+ Calibration
OOF custom metric	0.4090	0.4072	0.4099	0.4071	0.4071

- Ridge meta-weights: `lgb=0.229` , `cat=0.713` , `xgb=0.134` — CatBoost dominates the stack, consistent with it also having the best individual OOF score.
- Calibration shift ≈ 0 (the stacked OOF mean already matches the training target mean of 0.4780 to 8 decimal places) — confirming the calibration insight from §1.2 generalizes to the reduced ensemble.
- Top features by LightGBM gain: `district_te` (5,345), `inund_log1p` (2,340), `distance_to_river_m` (1,835), `reason_has_flood` (1,345), `extreme_x_terrain` (1,303) — `district_te` remains the dominant feature, as in every prior version.

The **0.4071** OOF score is higher (worse) than the Initial Round's best **0.383**, which is expected and accepted: that 0.383 came from a 5-model, 30-fold-equivalent, pseudo-labeled search not intended to run on every request. The Final Round's goal was not to win the leaderboard again, but to demonstrate that **the same modeling ideas, re-engineered for single-record serving, remain competitive** while becoming deployable — and the gap (0.383 → 0.407) is the documented, deliberate cost of that trade-off.

- 4. **AI Risk Advisor.** A new component with no Initial Round analogue: `src/llm_advisor.py` turns the score + top contributing features into a plain-language summary and recommended actions via **Claude (Claude-haiku-4-5)**, with a deterministic template fallback if `ANTHROPIC_API_KEY` is not set — satisfying the "AI-powered product" track and Rule 5's disclosure requirement.
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4. MLOps Practices

4.1 Deployment Strategy

- **Containerization:** a single `Dockerfile` (Python 3.11-slim base) installs the minimal runtime dependency set (`requirements-app.txt` — pandas, numpy, scikit-learn, lightgbm, catboost, xgboost, fastapi, uvicorn, anthropic; the heavier dev/training dependencies — optuna, mlflow, matplotlib, pytest — are kept in `requirements.txt` only) and copies `src/`, `app/`, and the trained `models/v1/` artifacts into the image. `.dockerignore` excludes the dataset, MLflow store, virtualenv, and tests from the build context.
- **API serving:** `uvicorn app.main:app` serves both the REST API and the static frontend (`app/static/`) from one process.
- **Cloud deployment:** `render.yaml` defines a Render **Docker web service** with a `/health` check and an optional `ANTHROPIC_API_KEY` environment variable. Render builds the `Dockerfile` directly — no separate build step or artifact registry is needed, since `models/v1/` ships inside the image.
- **Web deployment:** the same image serves `/` (prediction form) and `/dashboard` (monitoring dashboard), so judges reach the full product through one URL with no extra setup, per the "Deployment Link" requirement.

4.2 Monitoring Approach

- **Prediction logging:** every `POST /predict` call writes a row to a local SQLite database (`app/monitoring.py`) — full input JSON, output score and risk category, model version, latency in milliseconds, and the number of out-of-distribution flags raised.
- **Error monitoring:** `app/main.py` wraps the prediction path in a try/except that returns a `400` with the error detail rather than a bare `500`, so client-side errors (e.g. malformed input that slips past pydantic) are visible and debuggable rather than silent.
- **Performance monitoring:** `GET /monitoring/stats` aggregates total predictions, average score, average latency, a 10-bin score histogram, risk-category counts, and per-district counts (top 10) — rendered as charts on `/dashboard` (Chart.js), refreshed automatically every 15 seconds.
- **User feedback collection:** `POST /feedback` lets a user report whether a flood actually occurred and rate the prediction 1–5; feedback counts and average rating are surfaced on the dashboard alongside prediction metrics.
- **Drift / OOD flagging:** `src/data_validation.flag_out_of_distribution()` compares each request's raw inputs against the training data's expected ranges (`NUMERIC_BOUNDS`) and returns human-readable flags, both in the API response (shown to the end user) and in the monitoring log (so an operator can see if incoming traffic is drifting from the training distribution).

4.3 Version Control Practices

- **Git repository** (`flood-guard-ai/`) with a `.gitignore` excluding the virtualenv, caches, training-run logs (`catboost_info/`), and the runtime SQLite database (`*.db`) — but **including** `data/` (the training/test CSVs), `models/v1/` (the deployed artifacts, ~14 MB), and `mlruns/` (the MLflow tracking store, ~300 KB), so the repository is self-contained: cloning it and running `uvicorn app.main:app` serves the exact model described in this report with no separate data or artifact download step.

- **Model versioning:** `src/config.MODEL_VERSION = "v1"` determines `models/v1/`; a future retraining run that changes the feature set or hyperparameters would be saved as `models/v2/` alongside `v1`, with `metadata.json` in each directory recording the training timestamp, metrics, feature count, and Ridge weights for direct comparison.
 - **Experiment tracking:** every `python -m src.train` run creates a new MLflow run under the `flood-risk-production` experiment (`mlruns/`), logging all hyperparameters and the full metric set (per-fold, OOF per-model, stacked, and calibrated). `mlflow ui --backend-store-uri ./mlruns` gives a local UI for comparing runs.
 - **CI/CD** (`.github/workflows/ci.yml`, bonus deliverable): on every push/PR to `main`, GitHub Actions installs dependencies, runs the full `pytest` suite (feature-pipeline shape/dtype tests, inference sanity tests, and API tests via FastAPI's `TestClient`), and then builds the Docker image — so a broken feature pipeline, a broken model artifact, or a broken Dockerfile is caught before deployment.
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5. Conclusion & Future Work

FloodGuard AI demonstrates that the Initial Round's core modeling ideas — a gradient-boosted ensemble with Ridge meta-stacking and mean calibration, built on a rich district/geo/interaction feature set — can be re-engineered into a deployable, monitored, single-record-serving system without abandoning what made the original model competitive.

Future improvements (also covered in the presentation):

- Periodic retraining as new labeled flood events become available, versioned as `models/v2/`, `v3/`, ... and compared via the MLflow run history.
- Re-introducing a lightweight, request-time-feasible form of pseudo-labeling using the monitoring database's accumulated predictions + user feedback as a weak-label source.
- A model-drift alert (e.g. comparing the rolling distribution of incoming `flag_out_of_distribution` counts against a baseline) surfaced on the dashboard.
- Expanding the LLM Risk Advisor to support multiple languages (Sinhala/Tamil) for end users in the regions the model covers.